

Embedded Systems Programming

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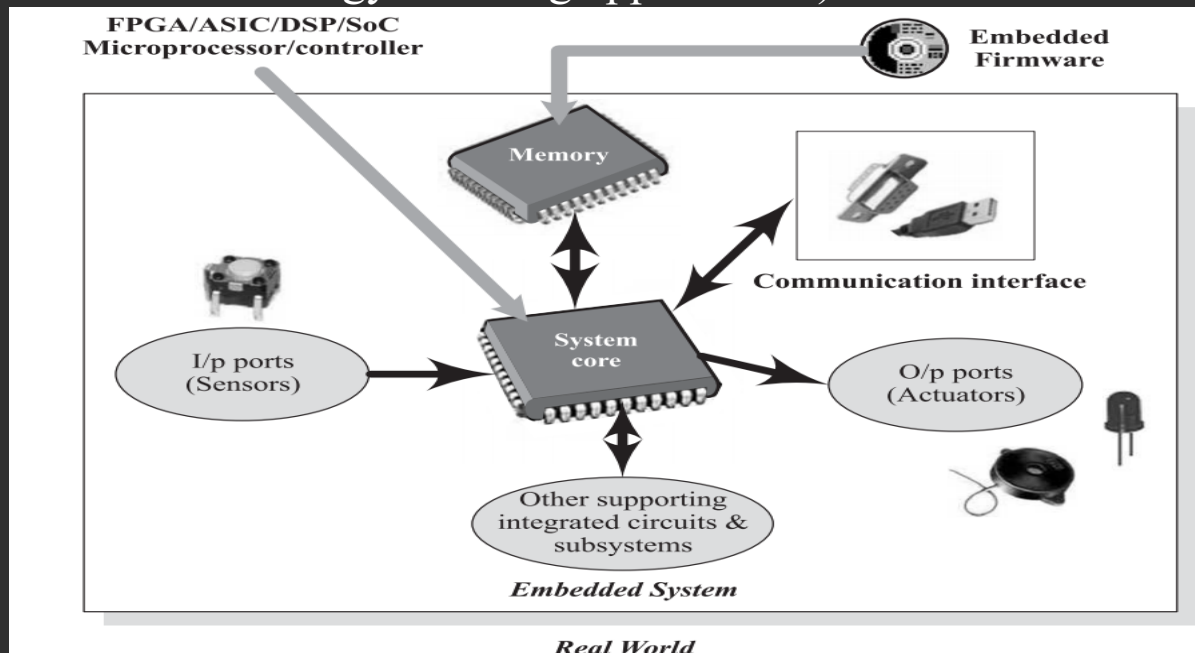
Lecture3



TYPICAL EMBEDDED SYSTEM

Overview

- ◆ A typical embedded system contains a single chip controller, which acts as the master brain of the system
- ◆ The controller can be:
 - ◆ a Microprocessor (e.g. Intel 8085) or
 - ◆ a microcontroller (e.g. Atmel AT89C51) or
 - ◆ a Field Programmable Gate Array (FPGA) device (e.g. Xilinx Spartan,A) or
 - ◆ a Digital Signal Processor (DSP) (e.g. Blackfi n® Processors from Analog Devices) or
 - ◆ an Application Specific Integrated Circuit (ASIC)/ Application Specific Standard Product (ASSP) (e.g. ADE7760 Single Phase Energy Metering IC from Analog Devices for energy metering applications)



Elements of an embedded system

CORE OF THE EMBEDDED SYTEM

- ◆ A typical embedded systems are domain and application specific and are built around a central core.
- ◆ The core of the embedded system falls into any one of the following categories:
 1. General Purpose and Domain Specific Processors
 - ◆ Microprocessors
 - ◆ Microcontrollers
 - ◆ Digital Signal Processors
 2. Application Specific Integrated Circuits (ASICs)
 3. Programmable Logic Devices (PLDs)
 4. Commercial off-the-shelf Components (COTS)
- ◆ Note that if you assess any embedded system, you will find that it is built around any of the core units mentioned above.

CORE OF THE EMBEDDED SYTEM



1. General Purpose and Domain Specific Processors
 - ◆ Almost 80% of the embedded systems are processor/controller based
 - ◆ A Microprocessor is a silicon chip representing a central processing unit (CPU), which is capable of performing arithmetic as well as logical operations according to a pre-defined set of instructions, which is specific to the manufacturer
 - ◆ Based on the purpose they serve; processors are classified as:
 - ◆ General-Purpose Processors (GPPs)
 - ◆ and Application-Specific Instruction Set Processors (ASIPs)
 - ◆ General Purpose Processor (GPP)
 - ◆ A GPP is a processor designed for general computational tasks (e.g., processors used in desktop and laptop computers). Due to mass-production their cost is low
 - ◆ Application-Specific Instruction Set Processor (ASIP):
 - ◆ ASIP is a processor with architecture and instruction set optimized for a specific application

CORE OF THE EMBEDDED SYTEM



- ◆ Based on the domain and application, the processor may be a microprocessor, microcontroller or a digital signal processor (DSP)
- ◆ Most of the applications use microprocessors or microcontrollers. Some speech and video signal processing applications use DSPs
 - ◆ Microprocessors
 - ◆ Examples: Zilog Z80 (8-bit), Intel 8086 (16-bit), Motorola 68020 (32-bit)
 - ◆ Raspberry pi
 - ◆ Microcontrollers
 - ◆ Microcontrollers are more powerful and more preferred than microprocessors. They may be either general-purpose microcontrollers (Intel 8051) or domain specific microcontrollers (Atmel's AVR). TMS1000 (from Texas instruments) is world's first microcontroller. Some of Arduino board microcontroller (ATMEGA328)
 - ◆ Digital Signal Processors (DSPs)
 - ◆ They are powerful, special-purpose, 8- or 16- or 32-bit microprocessors. They are used in signal processing applications. (e.g., Speech processing, image processing, video processing, video games, digital cameras. HDTV, radar, sonar, etc.)
 - ◆ They are 2 or 3 times faster than the general-purpose microprocessors
 - ◆ They involve large number of real-time calculations
 - ◆ They implement algorithms in hardware which speeds up the execution
 - ◆ Examples for DSP: Motorola 56001

CORE OF THE EMBEDDED SYTEM



◆ Classification of Processors/Controllers:

◆ Processors/controllers are classified according to the following criteria:

1. Instruction set
2. Architecture
3. Order of storing bits

1. Based on instruction set:

◆ RISC processors/Controllers:

- ◆ RISC stands for Reduced instruction Set Computing. RISC processors/controllers have a smaller number of instructions.
- ◆ Example: ARM, AVR microprocessors
- ◆ Important applications: Security systems, home automation, etc

◆ CISC processors / Controllers:

- ◆ CISC stands for Complex Instruction Set Computing. CISC processor / controller has large and complex instruction set.
- ◆ Example: VAX, AMD microprocessors
- ◆ Important applications: Smartphones, PDAs.

CORE OF THE EMBEDDED SYTEM

❖ Classification of Processors/Controllers:

1. Based on instruction set...

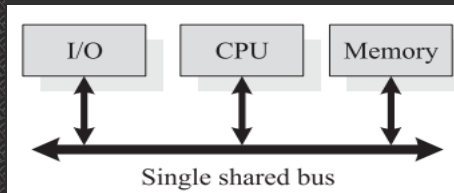
RISC	CISC
Lesser number of instructions	Greater number of Instructions
Instruction pipelining and increased execution speed	Generally no instruction pipelining feature
Orthogonal instruction set (Allows each instruction to operate on any register and use any addressing mode)	Non-orthogonal instruction set (All instructions are not allowed to operate on any register and use any addressing mode. It is instruction-specific)
Operations are performed on registers only, the only memory operations are load and store	Operations are performed on registers or memory depending on the instruction
A large number of registers are available	Limited number of general purpose registers
Programmer needs to write more code to execute a task since the instructions are simpler ones	Instructions are like macros in C language. A programmer can achieve the desired functionality with a single instruction which in turn provides the effect of using more simpler single instructions in RISC
Single, fixed length instructions	Variable length instructions
Less silicon usage and pin count	More silicon usage since more additional decoder logic is required to implement the complex instruction decoding.
With Harvard Architecture	Can be Harvard or Von-Neumann Architecture



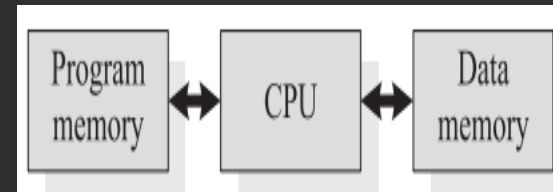
2. Architecture

- ◇ Von-Neumann architecture microprocessors/microcontrollers
 - ◇ It shares a single bus for program memory and data memory.
 - ◇ Processor first fetches instruction and then fetches data needed. Two separate fetches slow down processor operation.
 - ◇ Program memory and data memory are stored on same chip. So, there is a chance of accidental corruption of program memory.
 - ◇ Cheaper and no memory alignment problems
 - ◇ Von-Neumann architecture is also referred as Princeton architecture, since it was developed by the Princeton University.
- ◇ Harvard architecture microprocessors/microcontrollers
 - ◇ It contains separate buses for program memory and data memory. So, it has more speed.
 - ◇ Program memory and data memory are stored at different locations. So, no chances for accidental corruption of program memory
 - ◇ It is costly and has memory alignment problems

2. Architecture



Von-Neumann



Harvad

Harvard Architecture	Von-Neumann Architecture
Separate buses for instruction and data fetching	Single shared bus for instruction and data fetching
Easier to pipeline, so high performance can be achieved	Low performance compared to Harvard architecture
Comparatively high cost	Cheaper
No memory alignment problems	Allows self modifying codes [*]
Since data memory and program memory are stored physically in different locations, no chances for accidental corruption of program memory	Since data memory and program memory are stored physically in the same chip, chances for accidental corruption of program memory



2. Application Specific Integrated Circuits (ASICs)
 - ◆ ASIC is a microchip designed for a specific application
 - ◆ ASICs are pre-fabricated or customized for a special purpose
 - ◆ They are not reprogrammable
 - ◆ They consume small area and so reduce the system size.
 - ◆ On mass-production, ASICs are profitable
 - ◆ Most of the ASICs are proprietary products. So, their developers keep the internal details confidential
 - ◆ **Applications of ASICs:** Chips used for toys, handheld computers, cell phones and voice recorders
 - ◆ Example:
 - ◆ The ADE7760 Energy Meter ASIC developed by Analog Devices for Energy metering applications is a typical example for ASSP

CORE OF THE EMBEDDED SYTEM

3. Programmable Logic Devices (PLDs)

- ◆ Logic devices provide specific functions, including device-to-device interfacing, data communication, signal processing, data display, timing and control operations, and almost every other function a system must perform
- ◆ There are two types of logic devices-
 - ◆ Fixed logic devices and
 - ◆ Programmable logic devices
- ◆ Fixed logic devices contain permanent digital circuits.
 - ◆ They perform one or a set functions.
 - ◆ Once manufactured, they cannot be changed
 - ◆ Example
 - ◆ a logic gate or a multiplexer or a flip-flop performs a given logic function that is known at the time of device manufacture
- ◆ Programmable logic devices
 - ◆ Complex Programmable Logic Devices (CPLDs)
 - ◆ Example : [XC9500 CPLD](#)
 - ◆ and Field Programmable Gate Arrays(FPGAs)
 - example: [Xilinx FPGAs](#) , [Cyclone V SX SoC DE10-Standard](#)

CORE OF THE EMBEDDED SYTEM

4. Commercial off-the-shelf Components (COTS)

- ◆ A Commercial Off-the-Shelf (COTS) product is one which is used 'as-is'. COTS products are designed in such a way to provide easy integration and interoperability with existing system components
- ◆ COTS refer to hardware or software products that are ready-made and available for sale, lease, or license in the commercial market

- ◆ **Advantages of COTS:**

- ◆ Reduce development time.
- ◆ Readily available and up-to-date
- ◆ You can access reviews and advice from existing users



[W6300-EVB-Pico2](#)

- ◆ **Disadvantages of COTS:**

- ◆ Have some unwanted extra features
- ◆ Additional licensing cost
- ◆ Manufacturer of the product may withdraw the product or discontinue the production of COTS at any time
- ◆ Examples of COTS: MS-Office, Anti-virus software, Adobe Photoshop, etc

Further reading



- ◆ [Embedded Programming](#)
- ◆ [What Is An ASIP?](#)
- ◆ [Specialization Vs. Generalization In Processors](#)
- ◆ [Optimizing efficiency and flexibility in DSP systems](#)
- ◆ [Difference between DSP and General Purpose Processor](#)
- ◆ [Application and type of microprocessor](#)
- ◆ [Testing COTS Systems? Make Evaluation Count](#)
- ◆ [Commercial Off-the-Shelf Components \(COTS\) in Realizing Miniature Implantable Wireless Medical Devices: A Review](#)
- ◆ [What Is an Embedded System and Its Main Components](#)